

SENTHILNATHAN SELVARAJAN

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PROFESSIONAL EXPERIENCE

Software Engineer

Agnikul Cosmos Pvt. Limited

Chennai, Tamil Nadu, India

Jan. 2025 - Present

- Delivered and maintained **10+ ERP products** spanning manufacturing, production planning, task management, payroll, and mission-critical operations, serving **1,000+ concurrent users** across the organization.
- Improved application reliability by introducing **automated integration and regression test suites**, reducing production incidents and cutting manual QA time; optimized slow database queries and introduced caching at the API layer to reduce average response latency.
- Built a **remote sensing pipeline** (PyTorch, Sentinel-1/2 SAR) for tree density and species detection across AOIs, including a preprocessing stage for SAR image co-registration, tiling, and augmentation to handle variable AOI sizes.
- Fine-tuned **Mask R-CNN** for tree crown segmentation, achieving **>90% mAP** against FIA ground-truth data; iterated on anchor box configuration and loss weighting to improve small-object (sparse canopy) detection.
- Engineered a **document intelligence system** ingesting **1M+ documents** with **sub-500ms retrieval latency at p99**, using **QdrantDB** for vector indexing and **e5-mistral-7b-instruct** embeddings, maintaining approx. **99.99% uptime** across production workloads.
- Designed the **chunking and retrieval strategy** (semantic chunking, hybrid re-ranking) to balance recall and latency at scale; built a conversational interface on top of the retrieval system using LLMs, reducing information retrieval time by **80%**.

Software Engineer

Ailaysa Technologies Pvt. Limited

Chennai, Tamil Nadu, India

Jun. 2024 - Jan. 2025 (8 Months)

- Experimented with multiple e-book reader architectures to design a seamless migration path for native-language readers moving from paper to screens.
- Collaborated with the frontend engineering team to implement a **TypeScript-based native reader**, optimizing render performance and memory usage to **scale up to 500+ users**.
- Built a **PoC web-native book reader** with a custom reflow engine handling dynamic pagination across variable viewport sizes, supporting responsive typography and layout recalculation on resize.
- Implemented **EPUB/PDF parsing** at the content layer, extracting text, structure, and metadata to feed the reflow and pagination engine.

iOS Engineer (Intern)

IAgami Technologies Pvt. Limited

Trichy, Tamil Nadu, India

Mar. 2024 - Dec. 2024 (10 Months)

- Developed an **offline-first agricultural assessment app** for **5k+ farmers** in low-connectivity environments, with local-first data storage, conflict resolution, and background queue flushing for eventual sync.
- Designed the **sync conflict-resolution strategy** to handle concurrent edits from field agents without data loss, prioritizing the most recent verified assessment.
- Built a native **iOS LiDAR module (Swift, ARKit)** capturing 3D point cloud data of crops to infer canopy depth and volume.
- Trained an **ML model on canopy depth features** to generate yield predictions, validating outputs against manually recorded farmer yield data.

SKILLS

Languages & Core: Rust, Go, Zig, C++, Async Python, TypeScript/JavaScript

Distributed Systems: Raft, CRDTs, gRPC, Consensus Protocols

ML & AI: PyTorch, Mask R-CNN, RAG, pgvector, RayServe

Geospatial: GDAL, Rasterio, Sentinel-1/2 SAR, ARKit/LiDAR

Backend & APIs: FastAPI, Django, Protobuf, REST

Infra & Systems: Docker, Tokio, Redis, AWS

Data Layer: MySQL, PostgreSQL, MongoDB, QdrantDB

Observability: Prometheus, Grafana

PROJECTS

kvgo | *Go, Raft* gh/kvgo

- Implementing the **Raft consensus algorithm** from scratch including leader election, log replication, and commit-index advancement, for a distributed key-value store with strongly consistent reads/writes.
- Handling node failure and recovery through persistent logs, with work in progress on snapshotting to bound log growth over time.

kuttygrad | *Python, NumPy* gh/kuttygrad

- Built an autograd engine from scratch implementing **reverse-mode automatic differentiation** with a NumPy backend and dynamic computation graph.
- Implemented custom gradient functions for common ops (matmul, conv, activations) and verified correctness via numerical gradient checking.

dossier | *Python, QdrantDB, Frappe* gh/dossier

- Built a production-ready **RAG system** optimized for large-scale indexing and low-latency document retrieval.
- Designed a modular ingestion pipeline (chunking, embedding, indexing) and a hybrid retrieval layer combining dense vector search with keyword filtering for higher precision.

EDUCATION

Bachelor of Technology in Artificial Intelligence and Data Science CGPA: 9.2/10
Sri Shanmugha College of Engineering and Technology 2021 - 2025
Relevant Coursework: Machine Learning, Deep Learning, Distributed Systems, Data Structures & Algorithms, Computer Vision